

Course: Operating system

Subject code : 24BTDS145

PROJECT REPORT

KrishiRakshak: AI-Based Early Detection and Management of Crop Diseases and Pest Infestations

A farmer- and extension-worker-friendly crop-health surveillance system with image-based diagnosis, weather-based risk forecasting, pest-trap monitoring, hotspot mapping, expert validation and multilingual advisories

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Abstract

Plant diseases and insect pests are among the largest causes of crop loss, and in India farmers usually recognise them only after visible damage has spread. Extension officers cover large areas and laboratory diagnosis is slow, while the factors that drive outbreaks — weather, crop stage, variety, soil and local pest history — are rarely combined into actionable farm-level alerts. This project presents KrishiRakshak, a web-based crop-health system that runs in the Chrome browser on desktop and mobile. A farmer photographs a diseased leaf and a MobileNetV2 transfer-learning model, trained on 55,448 PlantVillage images of 39 classes, returns the top-3 diagnoses with confidence. The diagnosis is combined with a 7-day weather-based risk forecast from twelve agro-meteorological disease models (e.g. the Hutton criteria for late blight), adjusted for crop stage, variety resistance, drainage, irrigation and confirmed cases nearby. Pest-trap counts and IoT sensor readings are compared with economic threshold levels. Confirmed cases are clustered with DBSCAN to detect geospatial hotspots and warn nearby farms. Every AI diagnosis enters a prioritised validation queue where extension officers confirm, correct or refer it to a laboratory; confirmed images are used to retrain the model. Integrated Pest Management advisories — cultural, biological and, only as a last resort, chemical control with safe-use instructions — are generated in English, Hindi, Telugu, Tamil, Kannada and Marathi, with read-aloud audio. Officials receive a surveillance dashboard. The model achieves 97.01% accuracy and 99.82% top-3 accuracy on 8,318 unseen test images (macro F1 0.963).

Keywords: crop disease detection, deep learning, MobileNetV2, transfer learning, integrated pest management, disease forecasting, DBSCAN, geospatial surveillance, multilingual advisory, Flask.

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1. Introduction

1.1 Context and background

Agriculture supports a large share of India's population, and crop protection is central to farm income and food security. The Food and Agriculture Organization estimates that up to 40% of global crop production is lost to plant pests and diseases every year. Many of these losses are avoidable if a problem is identified early and managed correctly.

In practice, early detection is difficult. Symptoms such as small leaf spots, yellowing or insect eggs are easy to miss, and many diseases look alike. Village-level extension officers are responsible for thousands of farmers across large areas, and plant-health laboratories are usually located at district headquarters. As a result, farmers often rely on input dealers or neighbours for advice, which can lead to wrong diagnoses and unnecessary or excessive pesticide sprays.

At the same time, the information needed to predict outbreaks already exists: weather forecasts, the crop's growth stage, the variety's resistance, soil drainage and reports of the disease in neighbouring villages. Recent advances in deep learning, smartphones with good cameras and free weather APIs make it possible to combine these sources into a single decision-support tool.

1.2 Problem statement

Problem Statement Title: Early detection and management of crop diseases and pest infestations.

Farmers often recognise crop diseases or pest infestations only after visible damage has spread. Extension staff may cover large areas, while laboratory diagnosis and expert advice may not be immediately available. Weather, crop stage, variety, soil condition and local pest history influence risk, but these inputs are rarely combined into actionable farm-level alerts. Incorrect diagnosis may lead to delayed treatment, excessive or inappropriate pesticide use, increased cultivation cost, residue concerns and yield loss. The challenge is to provide timely, reliable and locally relevant detection, forecasting and management support.

Why is this project necessary?

- **Late detection:** damage has already spread by the time symptoms are obvious.
- **Limited expert reach:** few extension staff and laboratories for a very large number of farmers.
- **Scattered risk information:** weather, crop stage, variety, soil and pest history are not combined into alerts.
- **Wrong treatment:** misdiagnosis causes delayed control, wasted money, pesticide residues and resistance.
- **No surveillance picture:** officials lack real-time data to plan preventive interventions.
- **Language barrier:** advice is rarely available in the farmer's own language.

1.3 Objectives and key goals

The expected solution is a farmer- and extension-worker-friendly crop-health system. The objectives of this project are:

1. **Image-based identification:** diagnose diseases from a leaf photograph using a convolutional neural network.
2. **Pest-trap and sensor inputs:** record trap counts and IoT sensor data and raise alerts above the economic threshold level (ETL).
3. **Weather-based risk forecasting:** compute a 7-day disease and pest risk for each farm.
4. **Geospatial hotspot mapping:** automatically detect clusters of confirmed cases and warn nearby farms.
5. **Expert validation:** let extension officers confirm or correct each AI diagnosis and refer difficult cases to laboratories.
6. **Multilingual advisories:** provide Integrated Pest Management (IPM) advice with safe input use in English, Hindi, Telugu, Tamil, Kannada and Marathi.
7. **Follow-up monitoring:** schedule re-checks and escalate cases that are not improving.
8. **Learning from field confirmations:** retrain the model with expert-confirmed field images.
9. **Dashboards:** give agriculture officials surveillance indicators and charts.
10. **Accessibility:** run as a website in Chrome on both desktop and mobile phones.

Expected outcomes include earlier detection, reduced crop loss, more targeted pesticide use, faster extension response, improved surveillance coverage and better planning of preventive interventions.

1.4 Scope of the project

The system covers 17 crops in its risk and pest modules and 14 crops in the image classifier (apple, blueberry, cherry, maize, grape, orange, peach, bell pepper, potato, raspberry, soybean, squash, strawberry and tomato). It includes 7 trap-monitored pests (fall armyworm, pink bollworm, Helicoverpa, brinjal fruit and shoot borer, yellow stem borer, brown planthopper and whitefly), three user roles (farmer, extension officer and administrator/official) and six languages.

2. Background and Related Work

Hughes and Salathé (2015) released PlantVillage, an open dataset of more than 50,000 expertly labelled leaf images. Mohanty et al. (2016) showed that deep convolutional networks (AlexNet and GoogLeNet) trained on PlantVillage can reach 99.35% accuracy, but noted that accuracy drops considerably on images taken in different conditions — a key reason why expert validation is included in this project. MobileNetV2 (Sandler et al., 2018) uses depthwise-separable convolutions and inverted residual blocks, which makes it about an order of magnitude cheaper than GoogLeNet-class models and suitable for mobile and CPU deployment.

Weather-based disease forecasting has a long history in plant pathology. The Mills table (1944) relates the number of leaf-wet hours and temperature to apple-scab infection; the Hutton criteria, adopted in the UK in 2017, flag a late-blight risk period when the minimum temperature is at least 10 °C and relative humidity is at least 90% for six hours or more on two consecutive days. Integrated Pest Management (IPM) programmes in India use economic threshold levels (ETL) from pheromone-trap catches or field counts to decide when control is justified.

For spatial surveillance, density-based clustering (DBSCAN; Ester et al., 1996) is widely used to find clusters of events of arbitrary shape without fixing the number of clusters in advance. Existing mobile apps typically provide either photo identification or generic advisories; few combine diagnosis, farm-specific forecasting, expert validation, hotspot surveillance and continual learning in one workflow. This gap motivates KrishiRakshak.

3. Methodology

3.1 Approach

The problem was decomposed into three questions a farmer and an officer need answered: What is the problem on this crop? How likely is it to spread on this farm in the next days? What should be done, safely, and who should check it? The approach combines three sources of intelligence:

- **Computer vision** — a CNN classifier identifies the disease from a leaf image;
- **Agro-meteorological forecasting** — rule-based weather models, adjusted for farm factors and local history, estimate short-term risk;
- **Human expertise** — extension officers validate diagnoses, and their decisions feed back into the model and the outbreak map.

These are delivered through a three-tier web application (Figure 1): a browser front-end, a Flask server with six intelligence modules, and a data layer with a SQLite database, model store, knowledge base and external services.

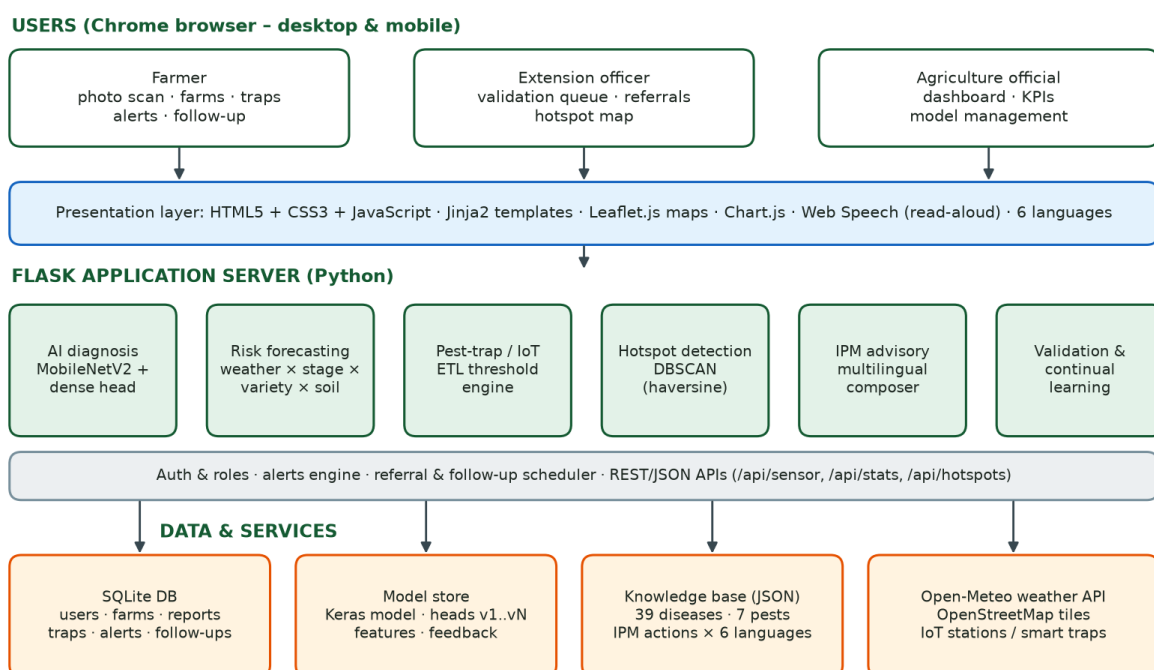


Figure 1: System architecture of KrishiRakshak

3.2 Tools and technologies

Table 1: Technology stack

Layer	Technology	Purpose
Front-end	HTML5, CSS3, JavaScript, Jinja2 templates	Responsive pages that run in Chrome on desktop and mobile
Maps	Leaflet.js 1.9, OpenStreetMap tiles, Leaflet.heat	Farm location picker, hotspot map and heatmap
Charts	Chart.js 4	Weather chart and officials' dashboard
Voice	Web Speech API	Reads advisories aloud in the selected language
Back-end	Python 3.12, Flask 3	Routing, authentication, REST APIs, alerts and scheduler
Deep learning	TensorFlow 2.18 / Keras 3, MobileNetV2	Leaf image classification by transfer learning
ML utilities	scikit-learn, NumPy, Pillow, Matplotlib	DBSCAN, metrics, image-quality checks, plots
Database	SQLite	Users, farms, reports, traps, sensors, alerts, model versions
Weather	Open-Meteo forecast API	Hourly temperature, humidity, rain and wind (past 3 + next 7 days)
IoT (simulated)	/api/sensor endpoint + iot_simulator.py	Weather-station and smart-trap input

Hardware used for development and training: Intel Core i7-10610U laptop CPU (4 cores), 16 GB RAM, no GPU, Windows 11.

3.3 Data collection and sample details

Image dataset

The PlantVillage dataset (version without augmentation, published on Mendeley Data) was used. It contains 55,448 colour images of single leaves on uniform backgrounds, in 39 classes: 38 crop–disease or crop–healthy classes of 14 crops and one background class (images without leaves). Class sizes range from 152 (potato healthy) to 5,507 (citrus greening) images (Figure 2). The data was split with stratified sampling into 70% training (38,813 images), 15% validation (8,317) and 15% test (8,318); the test set was used only once for final evaluation.

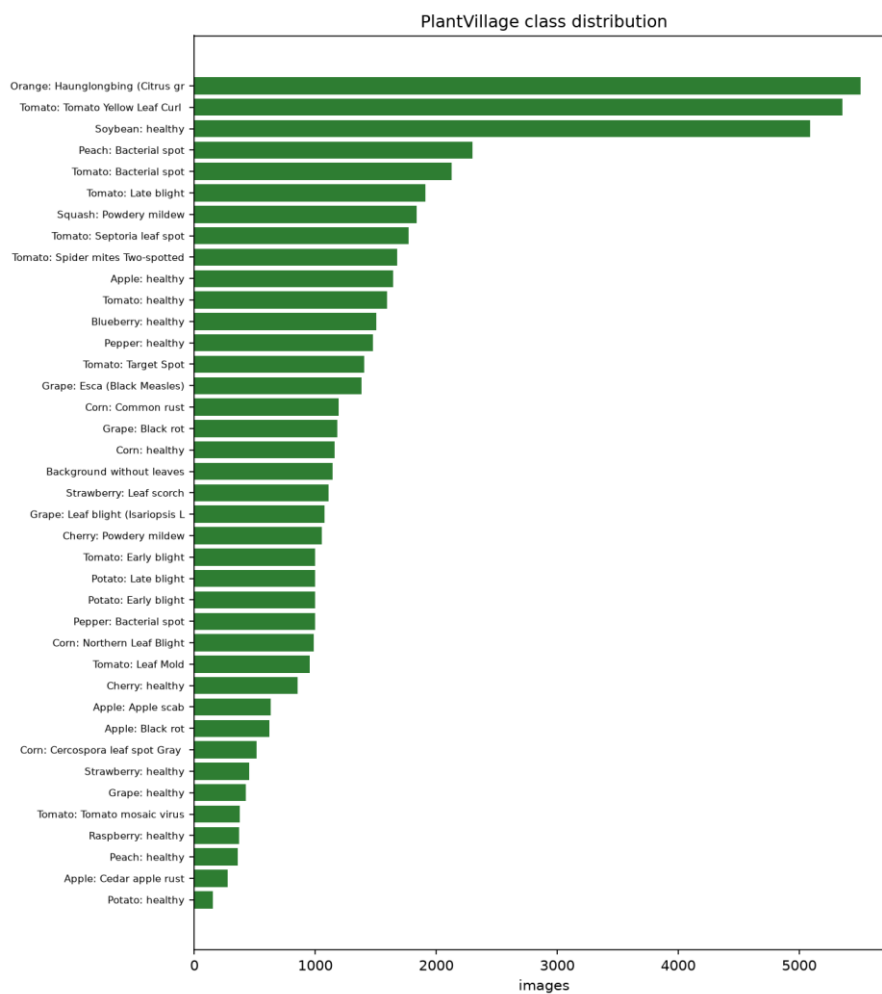


Figure 2: Number of images per class in the dataset

Weather data

For each farm, hourly temperature, relative humidity, precipitation and wind speed are fetched from the Open-Meteo API for the past 3 and next 7 days and aggregated into daily minimum/mean/maximum temperature, mean humidity, hours with $RH \geq 90\%$, leaf-wetness hours

(RH $\geq$ 90% or rain) and total rain. Results are cached for one hour; if the API is unreachable a seasonal sample is used and clearly labelled.

Knowledge base

A structured knowledge base was compiled from IPM package-of-practice literature: 39 disease/health profiles (pathogen, symptoms, spread speed, risk model, referral rule and IPM actions with doses and pre-harvest intervals), 7 pest profiles with trap type and ETL, 17 crop profiles, and a library of 47 IPM action phrases translated into six languages. The India-wide ban on agricultural use of streptomycin and tetracycline was respected by recommending copper-based bactericides and biological agents for bacterial diseases.

Field and demonstration data

Farm profiles, trap counts, sensor readings and reports are entered through the application. For evaluation, a demonstration database was generated with 107 farms in 32 villages of Maharashtra, Andhra Pradesh, Telangana, Tamil Nadu, Karnataka and Himachal Pradesh, 425 reports over 90 days and 960 trap readings. The images and the AI predictions in these reports are real (taken from the validation split and classified by the trained model), while farmers, locations, dates and outbreak clusters are synthetic. This allows the dashboard, hotspot and learning features to be evaluated end-to-end.

4. Process and Implementation

4.1 Detailed workflow

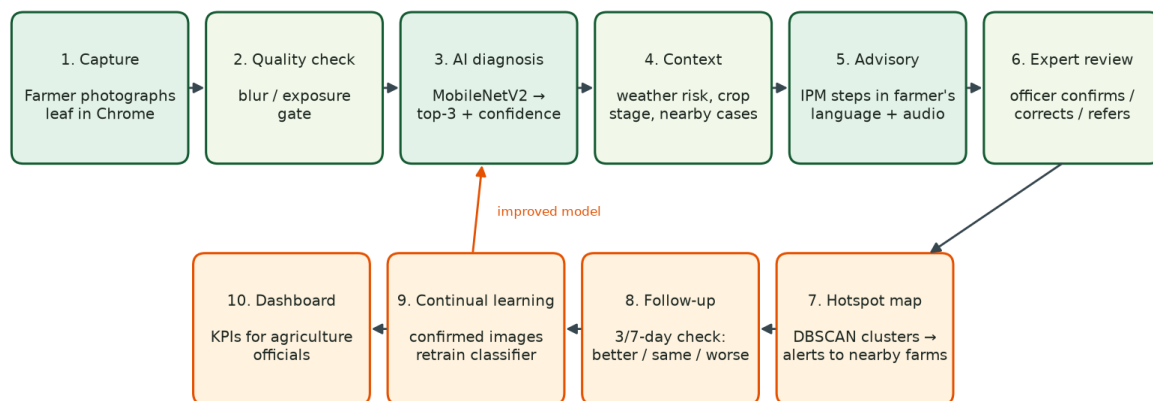


Figure 3: End-to-end workflow from photo capture to continual learning

- 1. Farm registration:** the farmer registers a farm with GPS location (map or phone GPS), crop, variety and its resistance, sowing date, area, soil type, drainage and irrigation method.
- 2. Daily risk alerts:** a background job (every 6 hours) computes the 7-day risk for every farm and creates alerts for high or severe risks; it also creates hotspot and follow-up alerts.
- 3. Photo capture:** in Chrome the farmer takes or uploads a leaf photo (camera capture on mobile, drag-and-drop on desktop) and selects the farm/crop.

4. **Quality gate:** the image is checked for blur and exposure; warnings are shown with the result.
5. **AI diagnosis:** the model returns the top-3 classes with confidence; crop-aware re-normalisation is applied and mismatches are flagged.
6. **Context and advisory:** the weather risk for the diagnosed disease on that farm is shown, together with the IPM advisory in the farmer's language and a read-aloud button.
7. **Referral:** if confidence is below 60%, or the disease is viral/notifiable (e.g. citrus greening), the nearest plant-health laboratory is assigned with directions.
8. **Expert validation:** the report enters the officer's queue, prioritised by notifiable disease, not-improving follow-ups, lab need, low confidence and fast spread. The officer confirms or corrects the label, sets severity and adds a note; the farmer is notified.
9. **Hotspot update:** confirmed cases are re-clustered and farms with the same crop near a hotspot are alerted.
10. **Follow-up:** a re-check is scheduled after 3 days (fast-spreading) or 7 days; outcomes 'same' or 'worse' create another follow-up and 'worse' escalates the case to officers.
11. **Continual learning:** confirmed images are stored as labelled examples and used to retrain the classifier.
12. **Dashboard:** officials monitor KPIs, trends, districts, AI confidence, follow-up outcomes and trap exceedances.

4.2 AI diagnosis model

Architecture. Each image is resized to 224×224 pixels and scaled to $[-1, 1]$. The ImageNet-pretrained MobileNetV2 backbone with global average pooling produces a 1280-dimensional feature vector. A classification head — Dropout(0.3) → Dense(256, ReLU) → Dropout(0.3) → Dense(39, softmax) — is trained on top.

Two-stage training. Because no GPU was available, the frozen backbone was run once over all images to cache their features (about 40 minutes at ~23 images/s). The head was then trained with the Adam optimiser (learning rate 10^{-3}), batch size 128, balanced class weights, early stopping on validation accuracy (patience 8) and learning-rate halving on plateau. Training stopped after 52 epochs and took 274 seconds. The backbone and head were then joined into a single Keras model for inference.

Crop-aware inference. When the farmer selects a crop, the class probabilities are restricted to that crop's classes plus the background class and re-normalised. If the unrestricted top class belongs to another crop with more than 60% probability, the farmer is asked to check the selected crop.

Continual learning. When an officer confirms or corrects a report, the stored 1280-d embedding of that image is saved with the final label. Retraining fine-tunes a copy of the current head for 6 epochs on the original training features plus the field examples (weighted ×5) at learning rate 2×10^{-4} . The new head version is activated only if its accuracy on the untouched test set does not fall by more than 0.5 percentage points, which protects the model against wrong labels. All versions are listed on the model page.

4.3 Weather-based risk forecasting

For each disease group, a daily favourability score between 0 and 1 is computed from the aggregated weather. Table 3 lists the main rules. The farm-level risk for the next seven days uses the highest three-day rolling mean of the daily scores and combines it with farm factors:

$$\text{Risk} = \min(100, 85 \times F_{\text{weather}} \times F_{\text{stage}} \times F_{\text{variety}} \times F_{\text{environment}} + \min(25, 6 \times N_{\text{nearby}}))$$

Table 2: Weather favourability rules (partial conditions score 0.45–0.55)

Model	Favourable conditions (score 1.0)	Crops
Late blight	Tmin ≥ 10 °C and ≥ 6 h RH ≥ 90%, Tmean ≤ 26 °C (Hutton criteria)	Potato, tomato
Early blight	Tmean 20–30 °C and ≥ 8 leaf-wet hours	Potato, tomato
Apple scab	Wet hours ≥ Mills-table requirement for the temperature (9–14 h)	Apple
Powdery mildew	Tmean 16–28 °C, RH 50–90%, rain < 2 mm (dry leaves)	Cherry, squash, grape, apple
Rust	Tmean 15–25 °C and ≥ 6 h RH ≥ 90%	Maize, soybean, apple
Bacterial spot	Tmean 24–32 °C with ≥ 5 mm splashing rain	Tomato, pepper, peach, brinjal
Black rot	Tmean 20–32 °C and ≥ 6 wet hours	Grape, apple
Northern leaf blight	Tmean 18–27 °C and ≥ 6 wet hours	Maize
Gray leaf spot	Tmean 22–30 °C and ≥ 12 h RH ≥ 90%	Maize
Leaf spots and moulds	Tmean 20–30 °C and ≥ 8 wet hours	Tomato, grape, strawberry, rice ...
Spider mites	Tmax ≥ 30 °C, RH < 50%, no rain	Tomato, pepper, brinjal
Whitefly/psyllid-borne virus	Tmean 26–35 °C, RH < 65%, rain < 2 mm	Tomato, cotton, orange

Table 3: Farm factors used in the risk score

Factor	Values
Crop stage (days after sowing ÷ crop duration)	seedling 0.85 · vegetative 1.0 · flowering/fruiting 1.15 · maturity 0.75 · harvested 0.4
Variety resistance	susceptible 1.0 · moderately resistant 0.75 · resistant 0.5
Drainage (humid diseases)	poor 1.15 · moderate 1.0 · good 0.92
Irrigation (humid diseases)	sprinkler 1.10 · flood 1.03 · rainfed 1.0 · drip 0.93
IoT sensor	leaf wetness ≥ 50% or RH ≥ 90% in the last 24 h adds +0.1 to humid-disease favourability
Local history	+6 per confirmed case (or trap exceedance) within 10 km in 21

	days, maximum +25
Risk level	Low < 25 ≤ Moderate < 50 ≤ High < 75 ≤ Severe

4.4 Pest-trap and sensor module

Trap readings are normalised to the pest's ETL unit: pheromone-trap counts are divided by the number of traps and the days since the last check (moths/trap/day or /week); visual counts are averaged per plant or leaf. A value at or above the ETL creates alerts for the farmer and the officers. IoT weather stations and smart traps post JSON to /api/sensor with a device key; `iot_simulator.py` demonstrates this.

Table 4: Pests and economic threshold levels (configurable)

Pest	Crop(s)	Monitoring	ETL used
Fall armyworm	Maize	Pheromone trap	3 moths/trap/day
Pink bollworm	Cotton	Pheromone trap	8 moths/trap/night (3 consecutive nights)
Fruit borer (Helicoverpa)	Tomato, Cotton	Pheromone trap	5 moths/trap/day
Brinjal fruit & shoot borer	Brinjal	Pheromone trap	5 moths/trap/week
Yellow stem borer	Rice (Paddy)	Pheromone trap	25 moths/trap/week
Brown planthopper	Rice (Paddy)	Field count	10 hoppers/hill
Whitefly	Cotton, Tomato, Brinjal	Yellow sticky trap / leaf count	8 adults/leaf

4.5 Hotspot detection

Reports from the last 30 days that are confirmed or corrected by an officer, or pending with AI confidence $\geq 90\%$, are grouped by disease. For each group DBSCAN is run with the haversine metric ($\epsilon = 5$ km, minimum 3 cases). Each cluster becomes a hotspot with centroid, radius, case count, cases in the last 7 days and severity (moderate ≥ 3 , high ≥ 6 , severe ≥ 12 cases or ≥ 6 in the last week). Farms growing the affected crop within the hotspot radius plus 5 km receive an alert. Farmers see hotspot areas only; officers also see individual reports, a heatmap and farm-level risk.

4.6 IPM advisory engine and multilingual support

Advisories follow the IPM hierarchy: (1) cultural/field practices, (2) biological control, (3) chemical control only as a last resort with product, dose and pre-harvest interval, (4) safe-use instructions (protective equipment, timing, label dose, rotation of chemical groups, container disposal), (5) referral and (6) follow-up. Each advisory is composed from the 47-phrasal action library, so every step is available in all six languages; chemical names and doses are inserted into translated templates. The whole user interface (338 strings) is translated, and the Web Speech API reads the

advisory aloud using the browser's Indian-language voices. Officers mark severity; low-severity cases are managed without chemical recommendations.

4.7 Database and application structure

Table 5: Database tables

Table	Main fields
users	username, password hash, role (farmer/officer/admin), name, phone, district, language
farms	owner, location (lat/lon), village, district, crop, variety, resistance, sowing date, area, soil, drainage, irrigation
reports	image, AI label/confidence/top-3, model version, quality notes, status, final label, severity, officer, referral, lab
followups	report, due date, status, outcome (better/same/worse), note
trap_readings	farm, pest, count, traps, days, normalised value, above-ETL flag, source (manual/sensor)
sensor_readings	farm, device, temperature, humidity, leaf wetness, soil moisture
alerts	user, farm, kind (weather/hotspot/trap/follow-up/validation), level, reference, read flag
labs, model_versions	referral laboratories/KVKs; model version history with accuracy

The code is organised as app.py (routes and APIs), risk.py (weather and risk models), hotspots.py (DBSCAN), ml_service.py (inference and retraining), kb/ (knowledge base), i18n/ (translations), templates/ (19 pages) and ml/train.py (training). Passwords are stored as salted hashes, pages are protected by role, farmers can only see their own farms and reports, and sensor devices authenticate with a device key.

4.8 Challenges and adaptations

Table 6: Challenges met during development

Challenge	Adaptation
No GPU on the development laptop	Cached frozen-backbone features: one CPU pass, then head training and retraining take seconds.
Strong class imbalance (152–5,507 images)	Balanced class weights; per-class precision/recall/F1 reported.
Laboratory images differ from field photos	Quality gate, crop-aware inference, top-3 display, low-confidence referral, expert validation.
Translating advice into six languages	Phrase library with templated doses so new diseases reuse existing translations.
Weather API outages	Hourly cache and a clearly labelled seasonal fallback.

Risk of wrong labels in retraining	Accept a new model only if held-out accuracy does not drop; keep version history.
Antibiotic ban for agricultural use	Copper-based and biological options recommended for bacterial diseases.

5. Results

5.1 Model performance

Table 7: Classification results on the held-out test set

Metric	Value
Test accuracy (top-1)	97.01%
Top-3 accuracy	99.82%
Macro-averaged F1-score	0.9631
Weighted F1-score	0.9700
Test images	8,318
Epochs (early stopping)	52
Head training time (CPU)	274 s
Model size (backbone + head)	13.7 MB

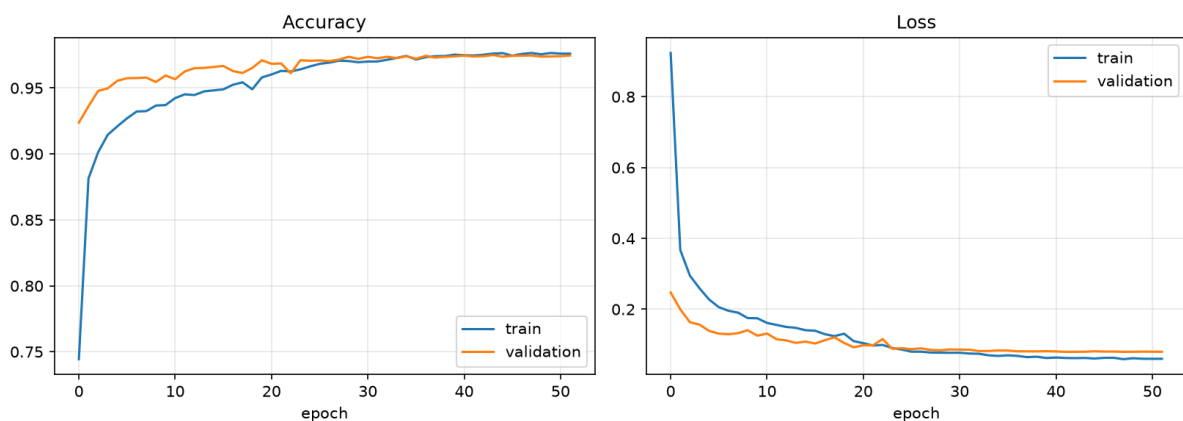


Figure 4: Training and validation accuracy and loss per epoch

Validation accuracy follows training accuracy closely and validation loss does not rise, showing that dropout and early stopping prevented over-fitting. (Training accuracy is slightly below validation accuracy because dropout is active during training.)

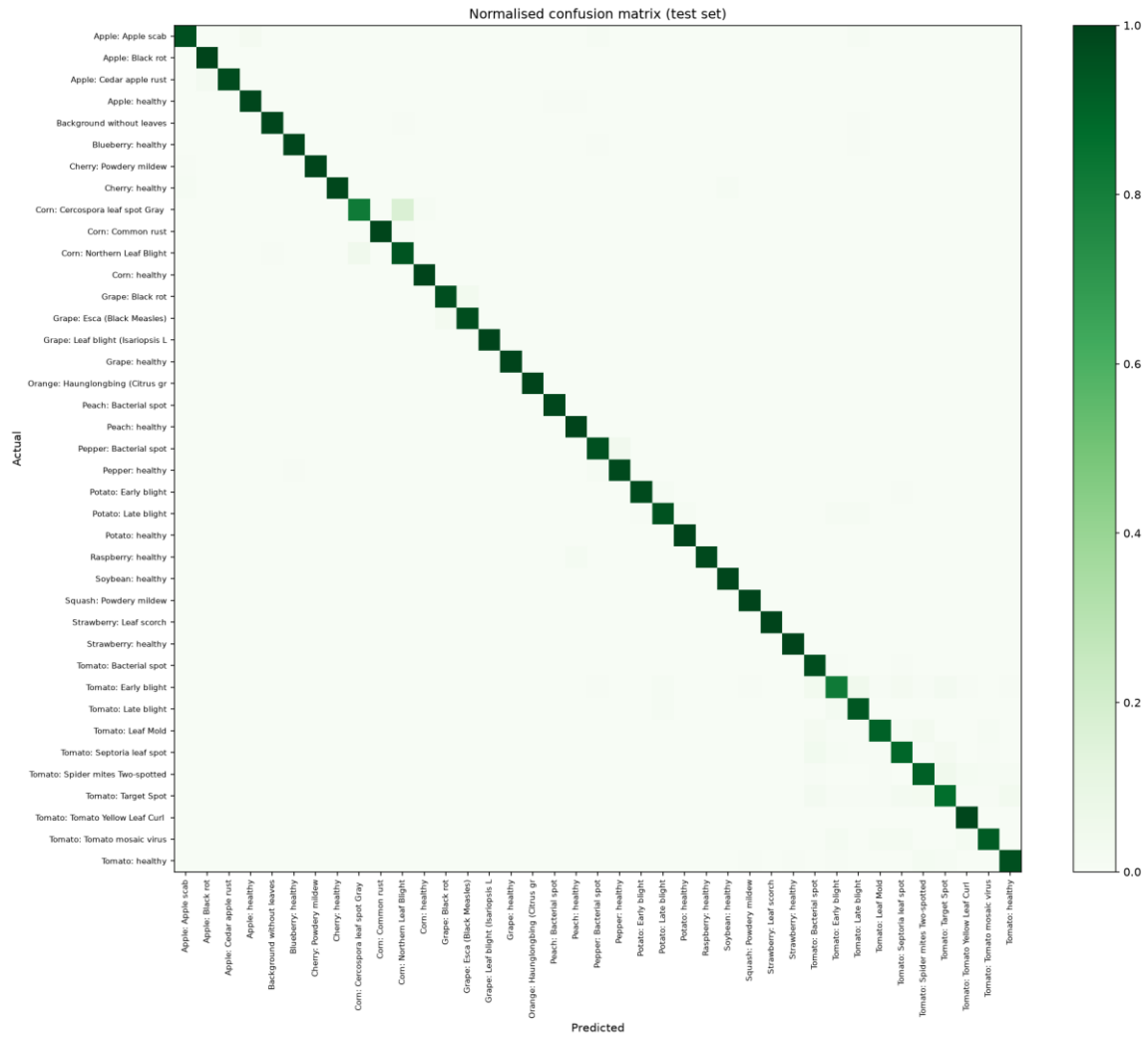


Figure 5: Normalised confusion matrix on the test set

Table 8: Per-class performance, weakest first

Class	Precision	Recall	F1	Test images
Tomato – Early blight	0.866	0.820	0.843	150
Maize – Gray leaf spot	0.900	0.818	0.857	77
Tomato – Target spot	0.867	0.867	0.867	211
Tomato – Mosaic virus	0.881	0.929	0.904	56
Tomato – Septoria leaf spot	0.941	0.891	0.915	266
Tomato – Two-spotted spider mite	0.927	0.913	0.920	252
Tomato – Leaf mould	0.935	0.909	0.922	143
Maize – Northern leaf blight	0.903	0.946	0.924	147
Tomato – Bacterial spot	0.912	0.972	0.941	319
Tomato – Late blight	0.938	0.944	0.941	287
Potato – Late blight	0.953	0.953	0.953	150
Tomato – Healthy	0.950	0.962	0.956	239

Capsicum – Bacterial spot	0.953	0.960	0.957	149
Potato – Healthy	0.920	1.000	0.958	23
Apple – Apple scab	0.968	0.957	0.963	94
Grape – Black rot	0.966	0.966	0.966	177
Grape – Esca (Black measles)	0.971	0.971	0.971	207
Capsicum – Healthy	0.960	0.982	0.971	222
Peach – Healthy	0.947	1.000	0.973	54
Strawberry – Healthy	0.971	1.000	0.986	68
Potato – Early blight	0.993	0.980	0.987	150
Apple – Cedar apple rust	1.000	0.976	0.988	41
Blueberry – Healthy	0.987	0.991	0.989	226
Apple – Black rot	0.979	1.000	0.989	93
Peach – Bacterial spot	0.994	0.986	0.990	345
Apple – Healthy	0.988	0.992	0.990	247
Tomato – Yellow leaf curl virus	0.989	0.991	0.990	804
Raspberry – Healthy	1.000	0.982	0.991	55
No leaf detected	0.994	0.988	0.991	171
Cherry – Healthy	1.000	0.984	0.992	128
Soybean – Healthy	0.999	0.990	0.994	764
Squash / Pumpkin – Powdery mildew	0.993	0.996	0.995	276
Cherry – Powdery mildew	1.000	0.994	0.997	158
Maize – Healthy	0.994	1.000	0.997	174
Maize – Common rust	1.000	0.994	0.997	179
Orange – Citrus greening (HLB)	0.999	0.999	0.999	826
Grape – Leaf blight (Isariopsis leaf spot)	1.000	1.000	1.000	161
Grape – Healthy	1.000	1.000	1.000	63
Strawberry – Leaf scorch	1.000	1.000	1.000	166

The weakest classes are Tomato – Early blight, Maize – Gray leaf spot, Tomato – Target spot. Most confusions occur between diseases of the same crop with similar lesions — for example maize gray leaf spot and northern leaf blight, or tomato target spot, septoria and early blight. Because the top-3 list almost always contains the correct class (99.82%), the officer can quickly correct such cases in the validation screen.

5.2 Application results

The complete application was run locally and tested in Google Chrome on desktop (1366×900) and at mobile width (390 px). The following screenshots show the main features.

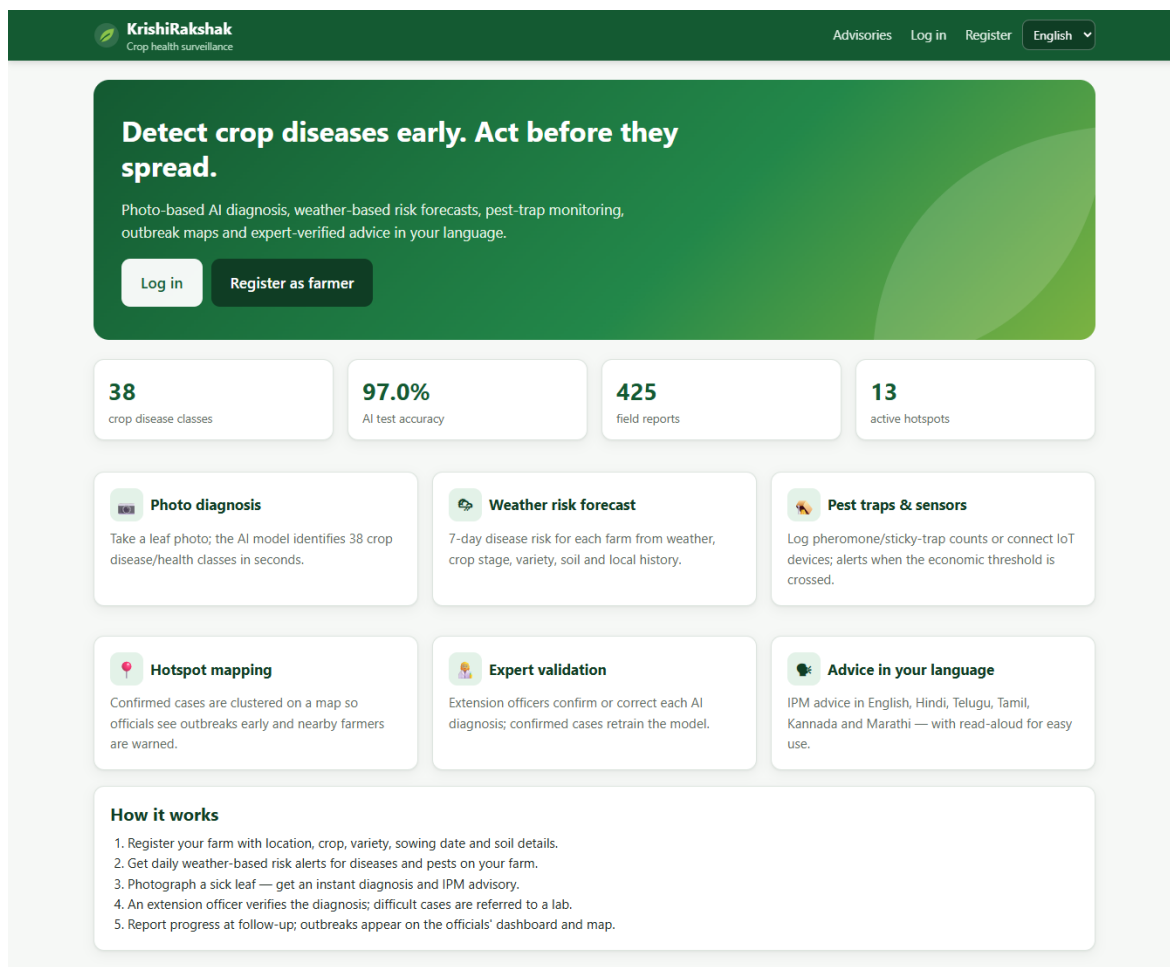


Figure 6: Landing page with feature overview

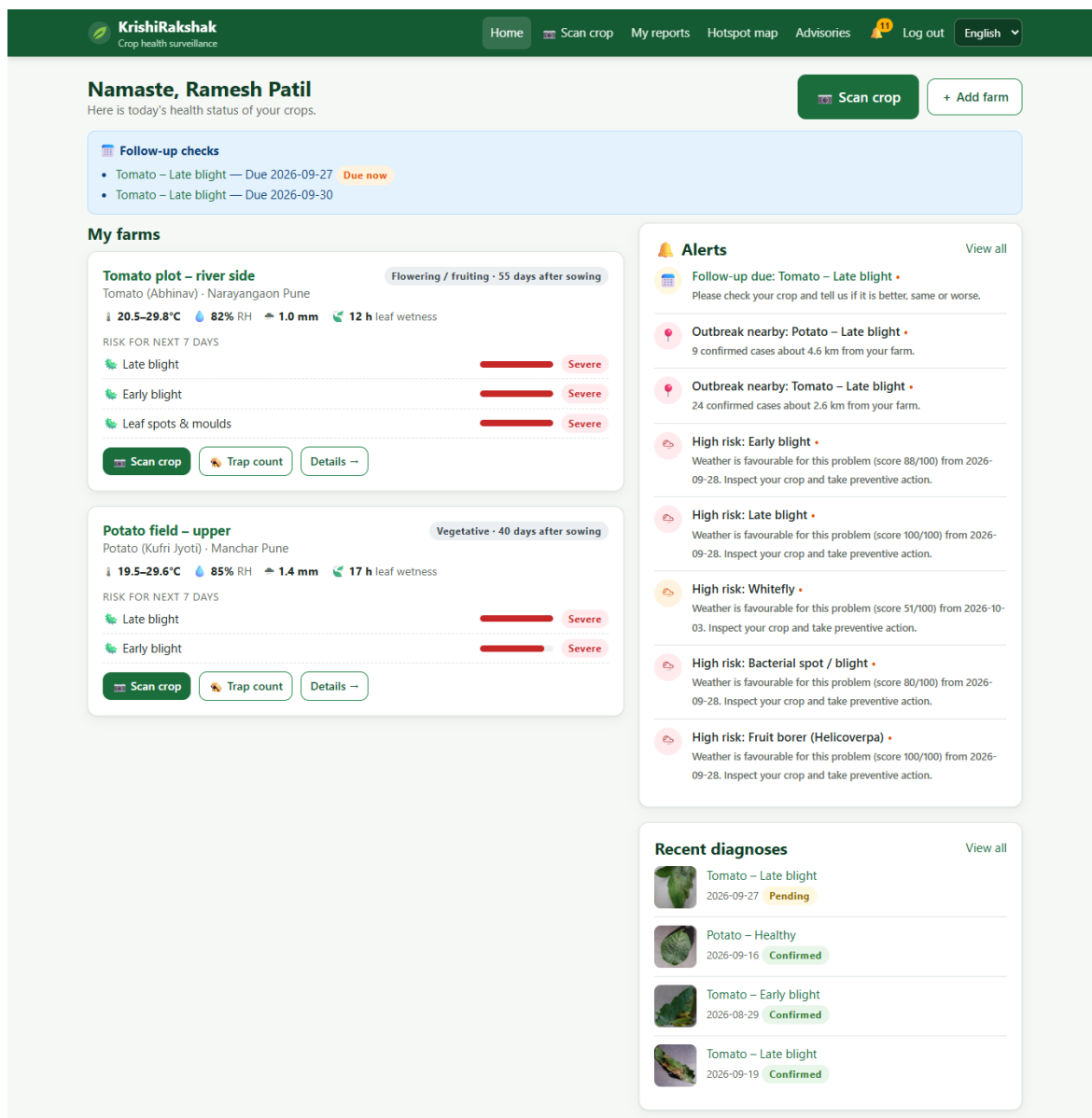


Figure 7: Farmer home: farms with live weather, 7-day risk, alerts and follow-ups

Tomato plot – river side

Tomato (Abhinav) · Narayangaon Pune · 19.115, 73.962

Scan crop

Flowering / fruiting

Crop stage · 55 days after sowing

Susceptible

Variety resistance

Poor

Drainage

Sprinkler

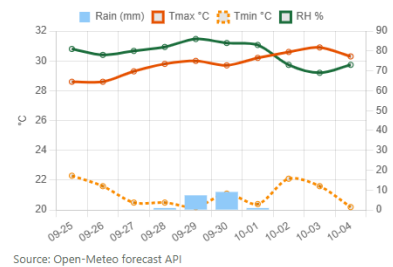
Irrigation

7-day disease & pest risk forecast

Score = weather favourability × crop-stage susceptibility × variety resistance × drainage/irrigation + confirmed cases within 10 km.

DISEASE / PEST	RISK	WEATHER	NEARBY CASES	PEAK FROM
Late blight	Severe 100	100%	20	2026-09-28
Early blight	Severe 100	100%	1	2026-09-28
Leaf spots & moulds	Severe 100	100%	0	2026-09-28
Fruit borer (Helicoverpa)	Severe 100	100%	3	2026-09-28
Bacterial spot / blight	Severe 80	65%	0	2026-09-28
Whitefly	High 51	30%	5	2026-10-03
Whitefly / psyllid-borne viruses	Moderate 44	45%	0	2026-10-01
Spider mites	Low 5	5%	0	2026-09-28

Weather (past 3 + next 7 days)



Pest trap log

Pest: Fruit borer (Helicoverpa) Insects counted:

Number of traps / plants: 1 Days since last check: 1

ETL: 5 moths/trap/day

Save

DATE	PEST	RATE	
2026-09-21 10:56	Whitefly	10.98 / ETL 8	Above ETL
2026-09-14 07:34	Whitefly	6.68 / ETL 8	OK
2026-09-07 06:44	Whitefly IoT	9.45 / ETL 8	Above ETL
2026-09-02 00:58	Whitefly	11.79 / ETL 8	Above ETL
2026-08-24 09:35	Whitefly IoT	4.72 / ETL 8	OK
2026-08-18 12:22	Whitefly	6.19 / ETL 8	OK
2026-08-10 13:25	Whitefly	7.42 / ETL 8	OK
2026-08-04 13:05	Whitefly IoT	4.41 / ETL 8	OK
2026-09-22 01:29	Fruit borer (Helicoverpa)	1.43 / ETL 5	OK
2026-09-14 17:43	Fruit borer (Helicoverpa)	3.57 / ETL 5	OK
2026-09-07 02:13	Fruit borer (Helicoverpa) IoT	1.57 / ETL 5	OK
2026-09-01 12:56	Fruit borer (Helicoverpa)	1.14 / ETL 5	OK
2026-08-24 17:32	Fruit borer (Helicoverpa)	1.71 / ETL 5	OK
2026-08-18 03:55	Fruit borer (Helicoverpa)	2.14 / ETL 5	OK
2026-08-10 06:04	Fruit borer (Helicoverpa) IoT	1.86 / ETL 5	OK

Field sensors (IoT)

DATE	°C	RH %	LEAF WETNESS %	SOIL MOISTURE %
2026-09-28 01:10	18.5	95.0	66.0	34.7
2026-09-28 00:10	19.6	94.0	86.0	33.9
2026-09-27 23:10	18.4	91.0	75.0	37.7
2026-09-27 22:10	17.9	92.0	68.0	34.8
2026-09-27 21:10	19.5	95.0	75.0	37.8
2026-09-27 20:10	17.4	92.0	61.0	34.0
2026-09-27 19:10	22.3	83.0	26.0	30.4
2026-09-27 18:10	23.3	84.0	13.0	32.6
2026-09-27 17:10	22.4	82.0	17.0	34.3
2026-09-27 16:10	25.8	76.0	14.0	37.4

► Device API format

Figure 8: Farm detail: risk forecast table, weather chart, pest-trap log and IoT sensor readings

KrishiRakshak
Crop health surveillance

Home Scan crop My reports Hotspot map Advisories Log out English

Scan a leaf

Tap to take or choose a photo
JPG or PNG, one leaf, clear focus

Farm
Tomato plot - river side - Tomato

Crop
Tomato

Selecting the crop improves accuracy.

For best results

- Photograph one affected leaf in daylight.
- Fill the frame with the leaf; avoid shadows.
- Hold the phone steady so the photo is sharp.

Diagnose

KrishiRakshak - AI-based early detection & management of crop diseases and pests

Figure 9: Leaf-scan page with camera capture, farm/crop selection and photo tips


KrishiRakshak
Crop health surveillance

[Home](#)
[Scan crop](#)
[My reports](#)
[Hotspot map](#)
[Advisories](#)


11

[Log out](#)

English



Report #425 · 2026-09-27 21:10 · Tomato plot – river side

Pending

Tomato – Late blight

Phytophthora infestans · Water-mould (oomycete) disease

AI diagnosis with **100.0%** confidence. An extension officer will verify it soon.

Top AI predictions

Tomato – Late blight	100.0%
Tomato – Bacterial spot	0.0%
Tomato – Leaf mould	0.0%


Weather risk for this disease on your farm is Severe (100/100) for the coming days. **Act now to stop the spread.**

Management advisory (IPM)

 Listen
 Print

Field practices

- Collect and destroy infected leaves and crop debris (burn or bury them).
- Ensure good field drainage and do not allow water to stagnate.
- Avoid overhead (sprinkler) irrigation; water at the base of plants in the morning.
- Keep the recommended plant spacing and prune to improve air circulation.

Biological control

- Spray *Trichoderma viride* 1% WP @ 5 g per litre of water.

Chemical control (last resort)

- Use chemical sprays only if the problem keeps spreading after cultural and biological measures.
- Spray Metalaxyl 8% + Mancozeb 64% WP @ 2.5 g/L. Do not harvest for 5 days after spraying.
- Spray Cymoxanil 8% + Mancozeb 64% WP @ 3 g/L. Do not harvest for 5 days after spraying.

Safe use of pesticides

- Wear gloves, mask, full-sleeve clothes and goggles while spraying.
- Spray in the early morning or evening; do not spray before rain or in strong wind.
- Use only the recommended dose and read the label; do not mix chemicals unless advised.
- Do not use the same chemical more than twice a season; rotate chemical groups to prevent resistance.
- Never reuse empty pesticide containers; wash the sprayer away from wells, ponds and streams.

Follow-up

- This problem spreads fast. Check the crop again after 3 days.

Doses are indicative. Always follow the product label and the advice of your local agriculture officer.

Follow-up

Due 2026-09-30

☐ Better
☐ Same
☐ Worse

Note (optional)

Save

Upload a new photo

Typical symptoms

Large greasy grey-green to brown blotches; white mould under leaves; firm brown rot on fruits.

Figure 10: Diagnosis result: top-3 predictions, weather-risk link and IPM advisory with read-aloud

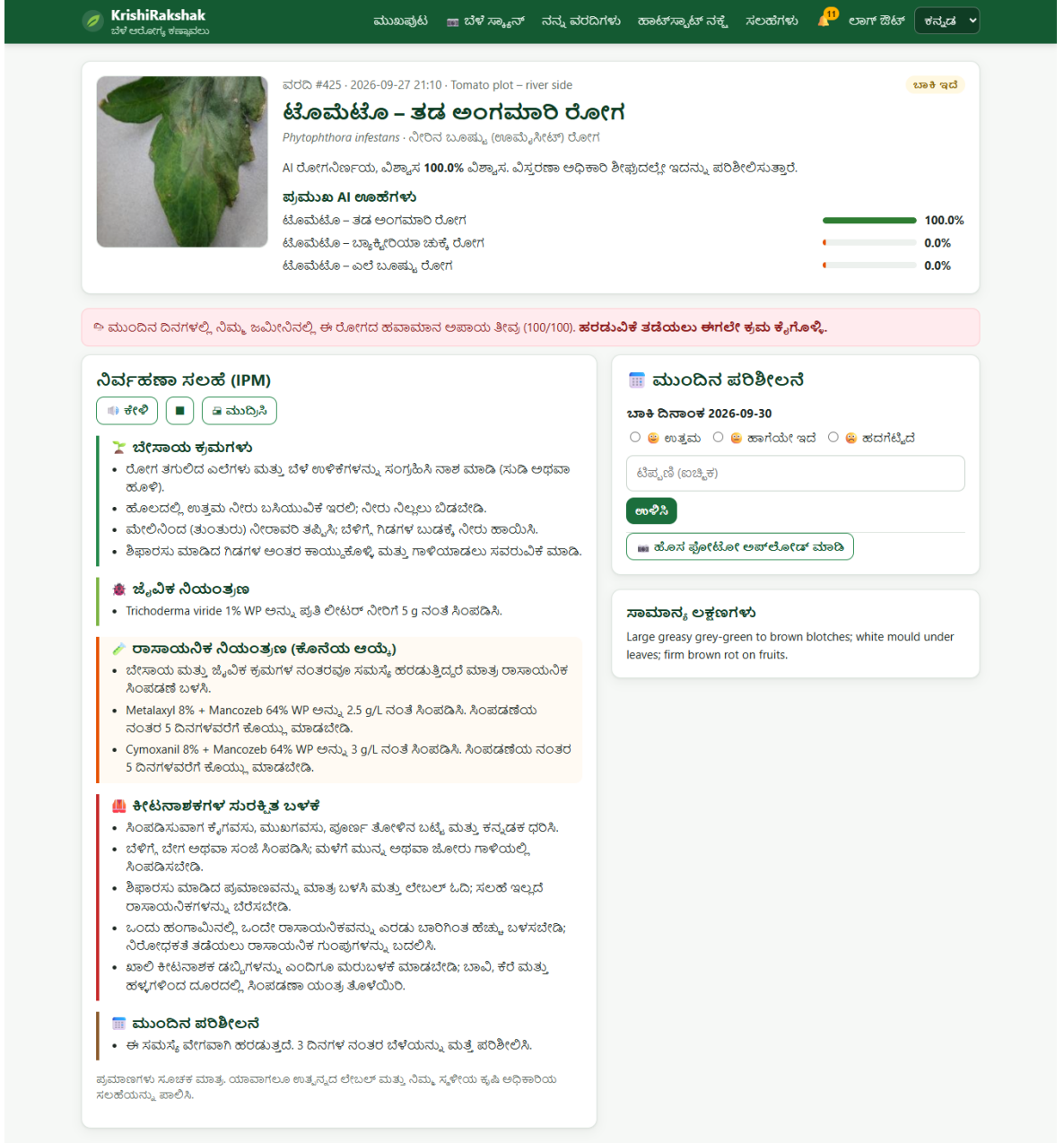


Figure 11: The same result and advisory in Kannada


KrishiRakshak
பயிர் நலக் கண்காணிப்பு

முகப்பு
பயிர் ஸ்கேன்
என் அறிக்கைகள்
பரவல் வரைபடம்
ஆலோசனைகள்
வெளியேறு
தமிழ்

← ஆலோசனைகள்

தக்காளி – பின் கருகல் நோய்

Phytophthora infestans - நீர்ப் பூஞ்சை (oomycete) நோய்

வழக்கமான அறிகுறிகள்: Large greasy grey-green to brown blotches; white mould under leaves; firm brown rot on fruits.

கேட்க

அச்சிடு

வயல் நடைமுறைகள்

- பாதிக்கப்பட்ட இலைகளையும் பயிர்க் கழிவுகளையும் சேகரித்து அழிக்கவும் (எரிக்கவும் அல்லது புதைக்கவும்).
- வயலில் நல்ல வடிகால் வசதி செய்யவும்; நீர் தேங்க விடாதீர்கள்.
- மேலிருந்து தெளிக்கும் (தெளிப்பு நீர்) பாசனத்தைத் தவிர்க்கவும்; காலையில் செடியின் அடியில் நீர் பாய்ச்சவும்.
- பரிந்துரைக்கப்பட்ட இடைவெளியில் நடவும்; காற்றோட்டத்துக்குக் கவாத்து செய்யவும்.

உயிரியல் கட்டுப்பாடு

- Trichoderma viride* 1% WP-ஐ ஒரு லிட்டர் நீருக்கு 5 g என்ற அளவில் தெளிக்கவும்.

இரசாயன கட்டுப்பாடு (கடைசி வழி)

- வயல் மற்றும் உயிரியல் முறைகளுக்குப் பிறகும் பிரச்சினை பரவினால் மட்டுமே இரசாயனத் தெளிப்பைப் பயன்படுத்தவும்.
- Metaxyl 8% + Mancozeb 64% WP-ஐ 2.5 g/L என்ற அளவில் தெளிக்கவும். தெளித்த பின் 5 நாட்களுக்கு அறுவடை செய்ய வேண்டாம்.
- Cymoxanil 8% + Mancozeb 64% WP-ஐ 3 g/L என்ற அளவில் தெளிக்கவும். தெளித்த பின் 5 நாட்களுக்கு அறுவடை செய்ய வேண்டாம்.

பூச்சிக்கொல்லிகளைப் பாதுகாப்பாகப் பயன்படுத்துதல்

- தெளிக்கும்போது கையுறை, முகக்கவசம், முழுக்கை ஆடை, கண்ணாடி அணியவும்.
- அதிகாலை அல்லது மாலையில் தெளிக்கவும்; மழைக்கு முன்போ பலத்த காற்றிலோ தெளிக்க வேண்டாம்.
- பரிந்துரைக்கப்பட்ட அளவை மட்டும் பயன்படுத்தி வேபிளைப் படிக்கவும்; ஆலோசனை இல்லாமல் மருந்துகளைக் கலக்க வேண்டாம்.
- ஒரு பருவத்தில் ஒரே மருந்தை இருமுறைக்கு மேல் பயன்படுத்த வேண்டாம்; எதிர்ப்புத் திறனைத் தடுக்க மருந்து வகைகளை மாற்றிப் பயன்படுத்தவும்.
- காலியான பூச்சிக்கொல்லி டப்பாக்களை மீண்டும் பயன்படுத்த வேண்டாம்; கிணறு, குளம், ஓடைகளிலிருந்து தொலைவில் தெளிப்பானைக் கழுவவும்.

பின்தொடர்தல்

- இந்தப் பிரச்சினை வேகமாகப் பரவும். 3 நாட்களுக்குப் பின் பயிரை மீண்டும் பார்க்கவும்.

அளவுகள் வழிகாட்டுதலுக்கு மட்டும். எப்போதும் மருந்து வேபிளையும் உங்கள் உள்ளூர் வேளாண் அலுவலரின் ஆலோசனையையும் பின்பற்றவும்.

Figure 12: Advisory page in Tamil

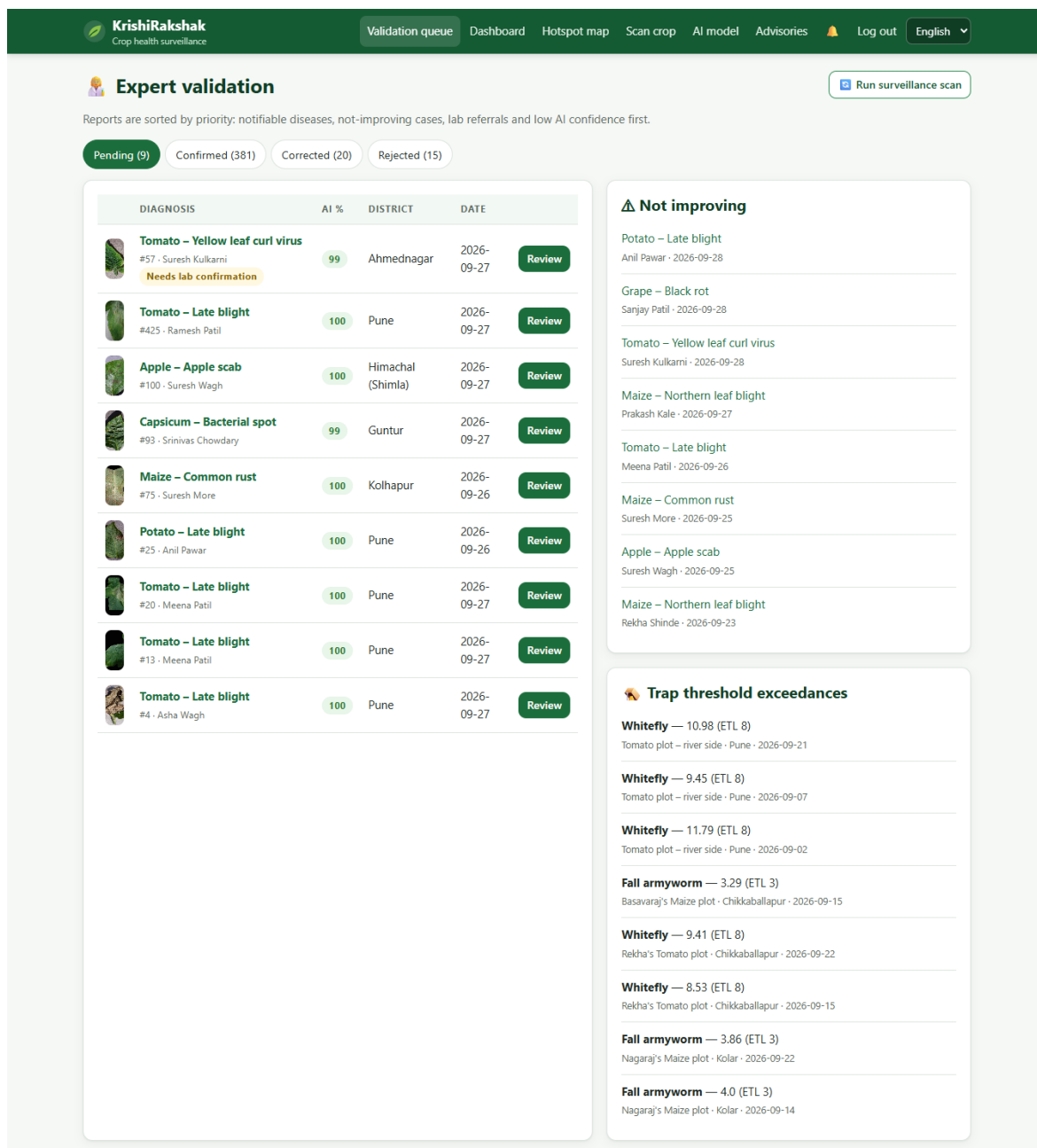


Figure 13: Extension officer's prioritised validation queue with trap and not-improving panels

KrishiRakshak
Crop health surveillance

[Validation queue](#)
[Dashboard](#)
[Hotspot map](#)
[Scan crop](#)
[AI model](#)
[Advisories](#)
[Log out](#)

English

← Validation queue

#57 · 2026-09-27 07:08

Pending

Tomato – Yellow leaf curl virus

Suresh Kulkarni
 Suresh's Tomato plot (Abhinav) · Sowing / transplanting date 2026-08-28
 Sangamner Ahmednagar · 19.5527, 74.2109

Top AI predictions

Tomato – Yellow leaf curl virus

99.3%

Tomato – Two-spotted spider mite

0.6%

Tomato – Leaf mould

0.1%

Needs lab confirmation

Follow-up
2026-09-30 — Pending

Expert decision
Confirmed diagnosis

Tomato – Yellow leaf curl virus

Keep the AI label to confirm, or pick another label to correct it.

Severity

Medium

Low severity = manage without chemical sprays.

☒ **Refer to laboratory**
District Plant Health Clinic, Ahmednagar (demo)

Note to farmer

☒ Open next pending report after saving

Confirm / correct

Reject

Confirmed and corrected images are added to the model's training data (continual learning).

Advisory the farmer sees

Field practices

- There is no chemical cure for viral diseases. Control the insects (whitefly/aphids) that spread the virus.
- Uproot and destroy infected plants immediately to stop the spread.
- Raise the nursery under a 40–50 mesh insect-proof net.
- Keep the field and bunds free of weeds and alternate host plants.

Biological control

- Install yellow sticky traps @ 10–12 per acre.
- Spray neem oil (1500 ppm) @ 5 ml per litre of water with a little soap as sticker.

Chemical control (last resort)

- Use chemical sprays only if the problem keeps spreading after cultural and biological measures.
- Spray Diafenthiuron 50% WP (whitefly control) @ 1.2 g/L. Do not harvest for 7 days after spraying.

Safe use of pesticides

- Wear gloves, mask, full-sleeve clothes and goggles while spraying.
- Spray in the early morning or evening; do not spray before rain or in strong wind.
- Use only the recommended dose and read the label; do not mix chemicals unless advised.
- Do not use the same chemical more than twice a season; rotate chemical groups to prevent resistance.
- Never reuse empty pesticide containers; wash the sprayer away from wells, ponds and streams.

Referral

- Send a sample to the nearest plant health laboratory for confirmation.
- Contact your extension officer or the nearest Krishi Vigyan Kendra.

Follow-up

- This problem spreads fast. Check the crop again after 3 days.

Doses are indicative. Always follow the product label and the advice of your local agriculture officer.

Figure 14: Expert review screen: confirm/correct, severity, lab referral and advisory preview

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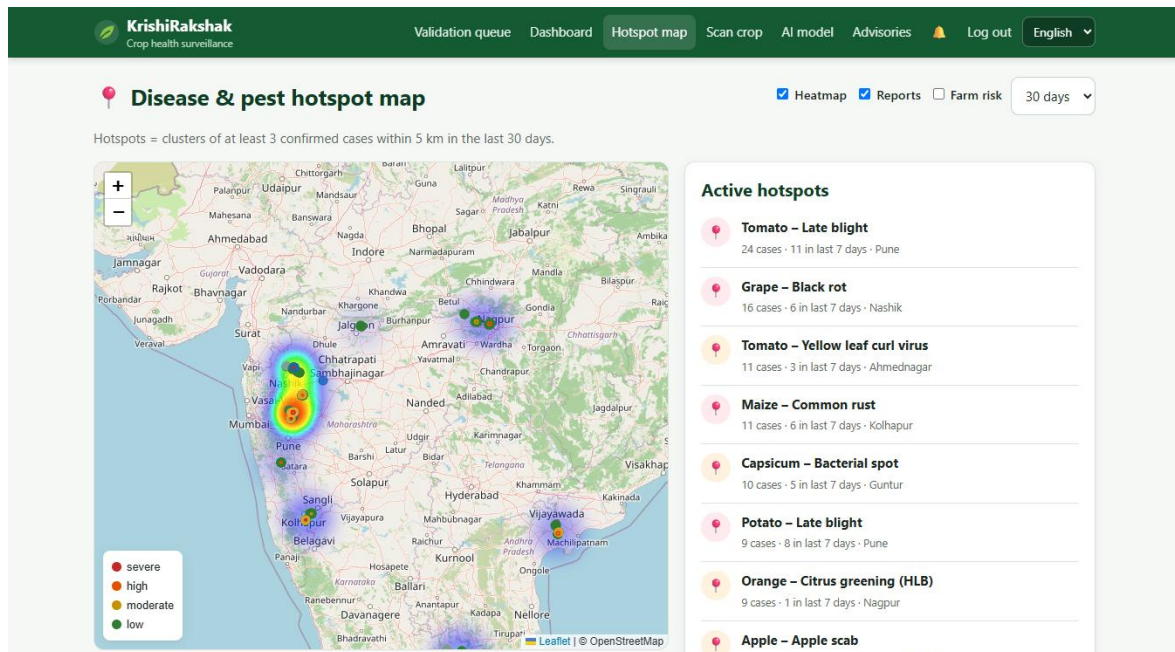


Figure 15: Hotspot map with DBSCAN clusters, report markers and heatmap

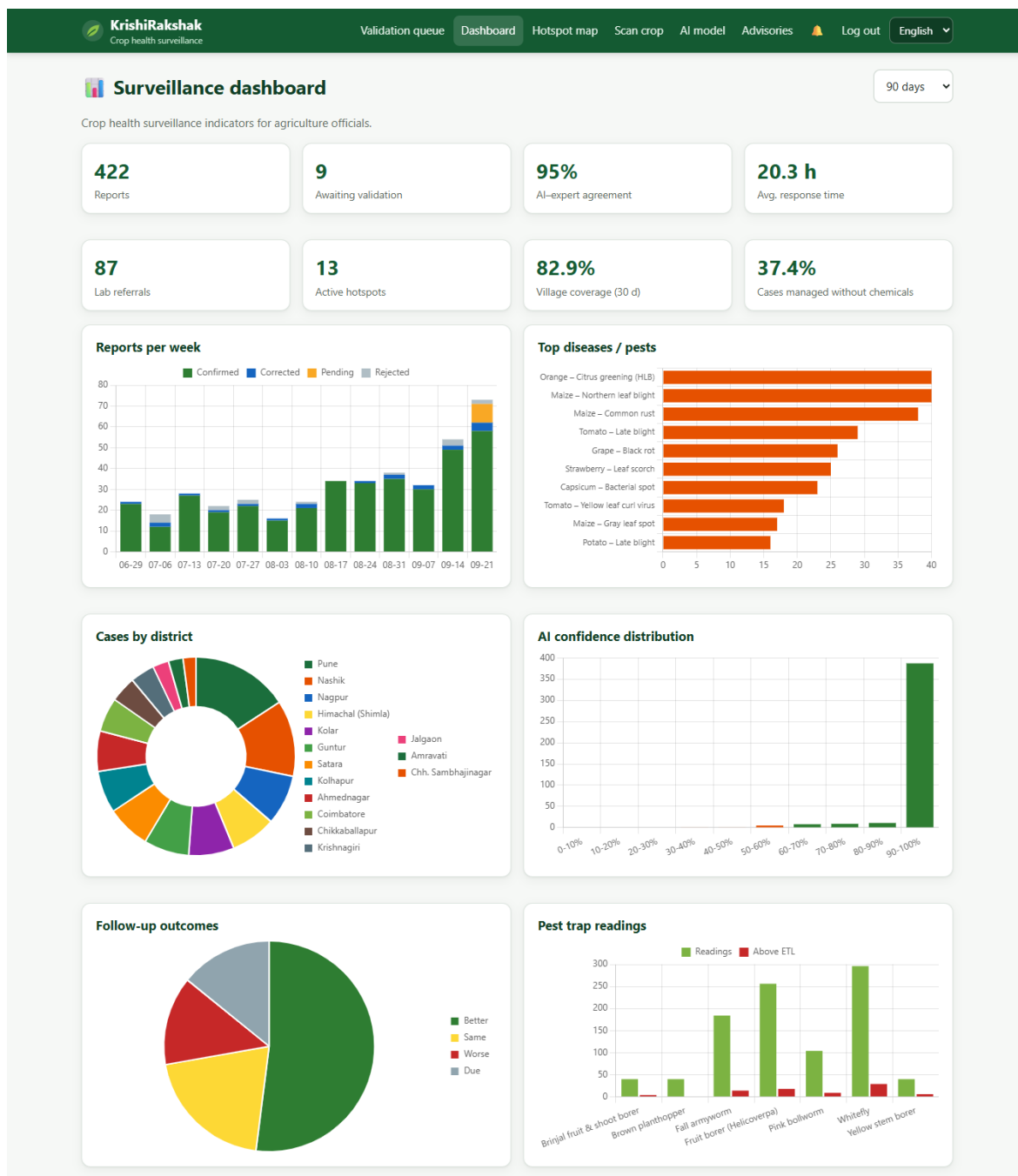


Figure 16: Surveillance dashboard for agriculture officials

AI model & continual learning

MobileNetV2 transfer-learning model trained on the PlantVillage dataset, improved with expert-confirmed field images.

97.01%

Test accuracy

99.82%

Top-3 accuracy

0.963

Macro F1

v2

Active

Training details

Architecture	MobileNetV2 (ImageNet, frozen) + Dense(256) head
Dataset	PlantVillage (Mendeley Data, without augmentation)
Images	55448 (39 classes)
Train / val / test	38813 / 8317 / 8318
Epochs	52
Trained on	2026-09-28 00:57

Learning from field confirmations

Every image an expert confirms or corrects is stored as a labelled example.

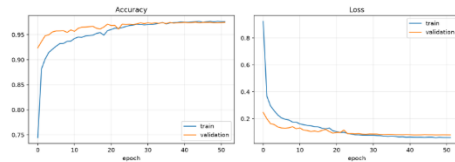
Retraining fine-tunes the classifier on these examples (weighted $\times 5$) plus the original data; the new version is activated only if accuracy on the held-out test set does not drop.

401 expert-confirmed field images available for learning

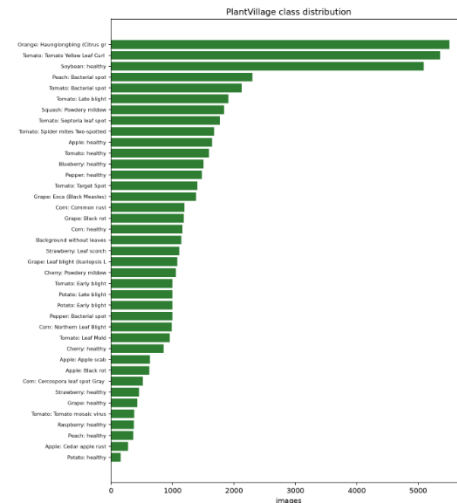
[Retrain with field data](#)

VERSION	TEST ACCURACY	FIELD ACCURACY	FIELD IMAGES	
v2	96.78%	99.8%	401	Active
v1	97.01%	—	0	initial PlantVillage model

Training curves



Class distribution



Confusion matrix

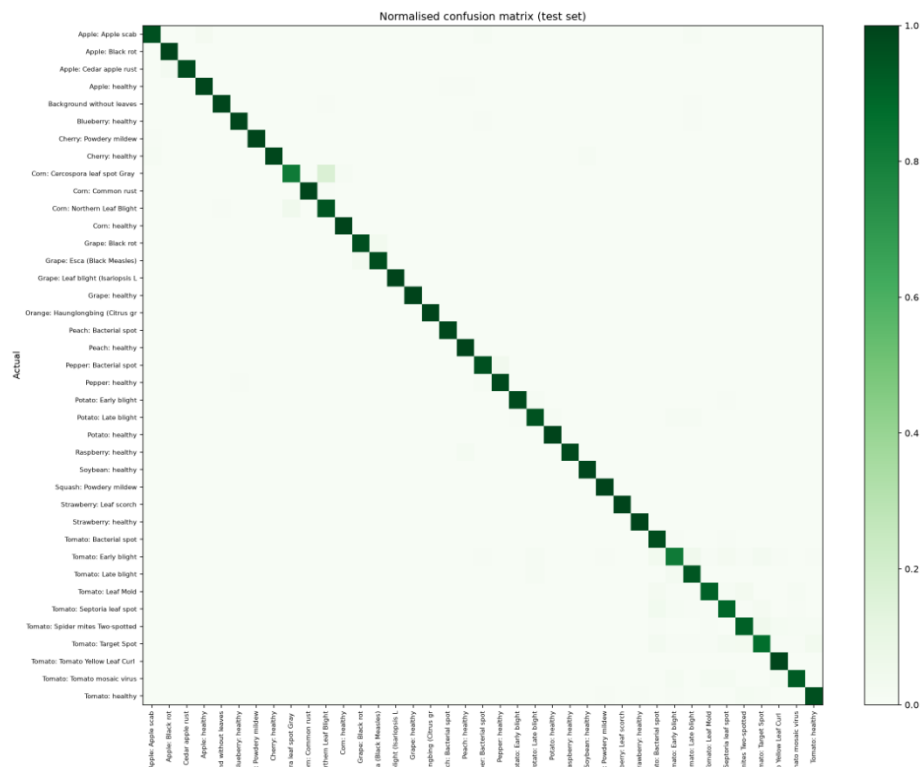


Figure 17: AI model page: metrics, version history and retraining from field data

 **KrishiRakshak**
Crop health surveillance



Namaste, Ramesh Patil

Here is today's health status of your crops.

Scan crop

+ Add farm

 **Follow-up checks**

- Tomato – Late blight — Due 2026-09-27 Due now
- Tomato – Late blight — Due 2026-09-30

My farms

Tomato plot – river side

Flowering / fruiting · 55 days after sowing

Tomato (Abhinav) · Narayangaon Pune

20.5–29.8°C

82% RH

1.0 mm

12 h leaf wetness

RISK FOR NEXT 7 DAYS

Late blight

Severe

Early blight

Severe

Leaf spots & moulds

Severe

Scan crop

Trap count

Details →

Potato field – upper

Vegetative · 40 days after sowing

Potato (Kufri Jyoti) · Manchar Pune

19.5–29.6°C

85% RH

1.4 mm

17 h leaf wetness

RISK FOR NEXT 7 DAYS

Late blight

Severe

Early blight

Severe

Scan crop

Trap count

Details →

 **Alerts** View all

 **Follow-up due: Tomato – Late blight** •
Please check your crop and tell us if it is better, same or worse.

 **Outbreak nearby: Potato – Late blight** •
9 confirmed cases about 4.6 km from your farm.

 **Outbreak nearby: Tomato – Late blight** •
24 confirmed cases about 2.6 km from your farm.

 **High risk: Early blight** •
Weather is favourable for this problem (score 88/100) from 2026-09-28. Inspect your crop and take preventive action.

 **High risk: Late blight** •
Weather is favourable for this problem (score 100/100) from 2026-09-28. Inspect your crop and take preventive action.

 **High risk: Whitefly** •
Weather is favourable for this problem (score 51/100) from 2026-10-03. Inspect your crop and take preventive action.

 **High risk: Bacterial spot / blight** •
Weather is favourable for this problem (score 80/100) from 2026-09-28. Inspect your crop and take preventive action.

 **High risk: Fruit borer (Helicoverpa)** •
Weather is favourable for this problem (score 100/100) from 2026-09-28. Inspect your crop and take preventive action.

Recent diagnoses View all

 Tomato – Late blight
2026-09-27 Pending

 Potato – Healthy
2026-09-16 Confirmed

 Tomato – Early blight
2026-08-29 Confirmed

 Tomato – Late blight
2026-09-19 Confirmed

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Figure 18: Farmer home at mobile-phone width

5.3 Surveillance indicators (demonstration data)

Table 9: Dashboard indicators computed from the demonstration database

Indicator (last 90 days)	Value
Reports received	422
Reports with disease/pest	379
Validated by officers	398
Pending validation	9
AI-expert agreement	95.0%
Average validation time	20.3 h
Laboratory referrals	87
Active hotspots detected	13
Village surveillance coverage (30 days)	82.9%
Confirmed cases managed without chemical advice	37.4%

The AI-expert agreement (95.0%) is measured on the real model predictions for the report images and is consistent with the test accuracy. The hotspot module detected 13 hotspots, including every simulated outbreak cluster: Tomato – Late blight (24 cases, Pune); Grape – Black rot (16 cases, Nashik); Tomato – Yellow leaf curl virus (11 cases, Ahmednagar); Maize – Common rust (11 cases, Kolhapur); Capsicum – Bacterial spot (10 cases, Guntur); Potato – Late blight (9 cases, Pune); Orange – Citrus greening (HLB) (9 cases, Nagpur); Apple – Apple scab (9 cases, Himachal (Shimla)); Maize – Northern leaf blight (9 cases, Coimbatore); Strawberry – Leaf scorch (8 cases, Satara); Tomato – Early blight (7 cases, Kolar); Orange – Citrus greening (HLB) (5 cases, Nagpur); Maize – Northern leaf blight (3 cases, Kolhapur).

5.4 Comparison with existing standards and approaches

Table 10: Comparison with published benchmarks and existing practice

Approach	Accuracy / capability	Notes
Mohanty et al. 2016 – GoogLeNet, fine-tuned	99.35% (PlantVillage, colour)	Full fine-tuning on GPU; ~6.8 M parameters
Mohanty et al. 2016 – AlexNet, fine-tuned	~99.3%	~60 M parameters
This project – MobileNetV2 frozen + head	97.01%	~2.6 M parameters; trained on a laptop CPU; retrain in seconds
Manual extension visit	expert-level	slow, limited coverage
Typical photo-ID apps	diagnosis only	no farm-specific forecasting, validation or surveillance

Our accuracy is a few points below fully fine-tuned networks because the backbone is frozen; in exchange the model is small, can be trained and retrained on ordinary hardware, and is combined with risk forecasting, expert validation and surveillance features that single-purpose tools lack.

6. Discussion

6.1 Interpretation of results

- A lightweight transfer-learning model reaches 97.01% accuracy, which is sufficient for a first-line screening tool, especially when combined with the top-3 list and expert validation.
- Linking the diagnosis to the farm's weather risk turns a label into a decision: the farmer sees whether conditions in the coming days favour spread and how urgently to act.
- Prioritising the officers' queue (notifiable, not improving, lab-needed and low-confidence cases first) makes better use of scarce extension staff.
- Ordering advice by IPM principles, attaching doses and pre-harvest intervals, and making chemical advice depend on severity supports more targeted pesticide use.
- Hotspot alerts give neighbouring farmers an early warning before they see symptoms, which directly addresses the problem of late detection.

6.2 Limitations

- PlantVillage images are taken on plain backgrounds under controlled conditions. Accuracy on real field photographs with complex backgrounds is expected to be lower until field images are collected and used for retraining.
- The classifier covers 14 crops; important Indian crops such as rice, cotton, chilli, pulses and onion have risk and pest modules but no image classes yet.
- The weather rules are simplified from published models and have not been calibrated against local outbreak records.
- Demonstration farms, reports and outbreaks are synthetic, and IoT devices are simulated.
- Doses in the knowledge base are indicative and must be checked against product labels and state recommendations.
- Live weather and map tiles need an internet connection; very low-literacy users may still need assistance.

6.3 Future work

- Collect and label field images (especially rice, cotton, chilli and pulses) and fine-tune the full network.
- Offline Android app with an on-device TensorFlow Lite model for areas with poor connectivity.
- SMS, WhatsApp and IVR voice alerts for farmers without smartphones; more Indian languages.
- Calibrate risk models with state pest-surveillance and weather-station data; add soil health card inputs.
- Integrate real LoRa/GSM smart traps with automatic insect counting by computer vision.

- Add satellite (NDVI) crop-stress layers and district-level forecasting of input demand.

7. Conclusion

KrishiRakshak shows that early detection and management of crop diseases and pests can be supported by a single, easy-to-use web system that runs in Chrome. It combines image-based symptom identification, pest-trap and sensor inputs, weather-based risk forecasting, geospatial hotspot mapping, expert validation, multilingual IPM advisories with safe-use guidance, laboratory referral, follow-up monitoring, learning from field confirmations and dashboards for agriculture officials — every element listed in the expected solution. With a test accuracy of 97.01% and an expert-in-the-loop design, the system can help farmers detect problems earlier, reduce crop loss and unnecessary pesticide use, and help extension services respond faster and plan preventive interventions.

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Appendix A: How to run the website

1. Install Python 3.12 and run: `pip install -r requirements.txt`
2. Train the model (downloads nothing else, uses ml/data/plantvillage): `python ml/train.py`

3. Create demonstration data: `python seed.py`
4. Start the server: `python app.py` (or double-click `run.bat`) and open `http://localhost:5000` in Google Chrome.
5. Demo logins are listed in `README.md` (farmer, officer and admin roles).
6. Simulate an IoT trap: `python iot_simulator.py --farm 1 --pest helicoverpa --count 7`

Appendix B: REST API

Table 11: Application programming interface

Endpoint	Method	Description
<code>/api/sensor</code>	POST	IoT weather station / smart trap data (header X-Device-Key)
<code>/api/farm/<id>/risk</code>	GET	7-day risk forecast for a farm
<code>/api/hotspots</code>	GET	Current DBSCAN hotspots
<code>/api/reports?days=N</code>	GET	Geo-referenced reports (officers)
<code>/api/farms-risk</code>	GET	Top risk for every farm (officers)
<code>/api/stats?days=N</code>	GET	Dashboard indicators and chart data (officers)